

Requirements Specification Document

1. Introduction

This project aims to build a healthcare insurance data pipeline using Big Data tools to clean, process, and analyze competitor and customer-related data so the company can improve revenue, understand customer behavior, customize insurance offers, and calculate royalties.

a. Purpose

This project aims to design and implement a healthcare insurance data pipeline using Big Data technologies to analyze competitor and customer-related data. The goal is to help the company improve revenue, understand customer behavior, customize insurance offers, and calculate royalties for policyholders.

b. Intended Audience and Use

This document is intended for Data Engineers, Data Analysts, Project Managers, and Business Stakeholders. It will guide development, testing, and deployment of the system.

c. Product Scope

The system will ingest healthcare datasets from multiple sources, clean and standardize data, store processed data in AWS Redshift, and generate insights based on business use cases.

Business Goals:

- Improve revenue
- Analyze customer behavior
- Identify profitable insurance groups
- Track claims and patient trends

d. Definitions and Acronyms

- S3: Amazon Simple Storage Service
- EMR: Elastic MapReduce
- ETL: Extract, Transform, Load
- PK: Primary Key
- FK: Foreign Key
- PySpark: Python API for Apache Spark
- Redshift: AWS Data Warehouse

2. Overall Description

a. User Needs

The system will be used by:

- Analysts to generate reports
- Business team to make decisions
- Engineers to maintain pipelines

Users need:

- Clean and reliable data
- Fast query performance
- Structured datasets for reporting

b. Assumptions and Dependencies

- Data will be available in CSV format
- AWS services (S3, Redshift, EMR/Databricks) are accessible
- PySpark will be used for processing
- Internet connectivity is required
- System depends on correct schema design

3. System Features and Requirements

a. Functional Requirements

- Disease with maximum claims
- Subscribers below age 30
- Group with maximum subgroups
- Hospital serving most patients
- Most subscribed subgroup
- Rejected claims count
- City with most claims
- Government vs private subscriptions
- Average premium
- Most profitable group
- Patients below 18 with cancer
- Cashless high charge patients
- Female 40+ knee surgery patients

b. External Interface Requirements

User Interface: SQL queries, Databricks/EMR notebooks

Hardware Interface: Laptop/PC, AWS Cloud

Software Interface: AWS S3, Redshift, PySpark

Communication Interface: Internet, AWS Console

c. System Features

- Data ingestion from S3
- Data transformation using PySpark
- Data storage in Redshift
- Query-based analytics
- Reporting-ready datasets

d. Nonfunctional Requirements

- Performance: Efficient handling of large datasets
- Security: Controlled access via AWS IAM
- Scalability: Cloud scalability
- Usability: Easy SQL querying
- Reliability: Stable data pipelines